

MASS-ENG-009

AI Orchestration Engine Specification

Page 1 – Purpose & Scope

Purpose

Define the enterprise AI orchestration subsystem responsible for coordinating AI providers, context, tools, and governance across MASS HQ.

Objectives

- Provide a unified AI interface
- Separate orchestration from AI providers
- Govern prompts, context, and tool execution
- Record all AI interactions for auditability

Out of Scope

Provider-specific model implementation and business workflow logic.

Page 2 – Architecture

Core Components

Provider Manager, Prompt Registry, Context Manager, Tool Invocation Service, Conversation Manager, Policy Guardrails, Audit Logger.

Relationships

Consumes Enterprise Core, Identity, Security, Event Bus, Knowledge Engine, and Document Engine. Serves all enterprise modules requiring AI capabilities.

Page 3 – Functional Specification

Requirements

1. Route requests to approved AI providers.
2. Assemble enterprise context securely.
3. Execute approved tools.
4. Enforce AI guardrails and policies.
5. Record prompts, responses, and execution metadata.

Non-Functional

Provider abstraction, extensibility, observability, deterministic orchestration, secure context isolation.

Page 4 – Interfaces & Examples

Interfaces

SubmitPrompt(), RetrieveContext(), InvokeTool(), SelectProvider(), RecordInteraction().

Example Flow

Workflow requests AI assistance → Context retrieved from Knowledge Engine → Guardrails applied → Provider selected → Response returned → Audit event published.

Page 5 – Construction Package

Claude Checklist

Implement provider abstraction; prompt registry; context assembly; tool execution framework; guardrail enforcement; audit logging; event integration; automated tests.

Definition of Done

AI requests execute through a governed orchestration layer with provider independence, audited

interactions, secure context handling, and standardized interfaces.

Constitution References

Knowledge, Security, Governance, Platform Principles, AI Governance.